

Derbyshire Shared Care Pathology Guidelines

Menopause

The menopause is defined as physiological ovarian failure, which typically occurs around the age of 50 years, but may be considered from 40 years onwards. It is associated with the cessation of menstruation and symptoms of oestrogen deficiency (low serum oestradiol).

Although serum oestrogen levels gradually fall as ovarian function declines, oestradiol levels are very variable in this situation and should not be measured to diagnose menopause. Falling oestradiol levels cause the gonadotrophins, follicle stimulating hormone (FSH) and luteinising hormone (LH) to rise significantly, particularly FSH.

In otherwise healthy women aged over 45yr with menopausal symptoms, a blood test is unnecessary for diagnosis.

Diagnose menopause without a blood test in these women when:

- No period for at least 12 months and not using hormonal contraception
- Based on symptoms alone in women without a uterus

Consider using FSH* to diagnose early menopause:

- In women aged 40-45yr with menopausal symptoms, including a change in their menstrual cycle

Peri-menopause

During peri-menopause, ovulation and menstruation occur at irregular intervals and this may occur over several years before menopause. FSH levels fluctuate greatly during this period and values may rise to post-menopausal values during individual non-ovulatory cycles. FSH levels can still be helpful in this context if they avoid the need for further investigations for an alternative cause of symptoms.

Diagnose peri-menopause without a blood test in otherwise healthy women over 45yr based on vasomotor symptoms and irregular periods

Do **not** use the following laboratory and imaging tests to identify perimenopause or menopause in people aged 45 or over:

Anti-Müllerian hormone(AMH);
Inhibin A; inhibin B;
Oestradiol;
Antral follicle count;
Ovarian volume.

These investigations may be requested in select patients on an individual basis e.g. if fertility investigations are required. This would typically be done in secondary care or in the fertility clinic. They are not used to diagnose Menopause.

Premature Ovarian Insufficiency

Premature ovarian insufficiency [POI] is defined as loss of ovarian activity under the age of 40 yrs and is characterised by amenorrhoea or irregular menstrual cycles with elevated gonadotrophins and low oestradiol. There are various causes of POI, including surgical oophorectomy, genetic defects, ovarian toxins, and autoimmune injury. It may be idiopathic and/or familial.

In younger women presenting with possible POI (< 40 years), measurement of FSH* can be useful. FSH >25 IU/L on two occasions taken 4 to 6 weeks apart suggests ovarian insufficiency.

Untreated, women with POI are at increased risk of cardiovascular disease, osteoporosis and dementia.

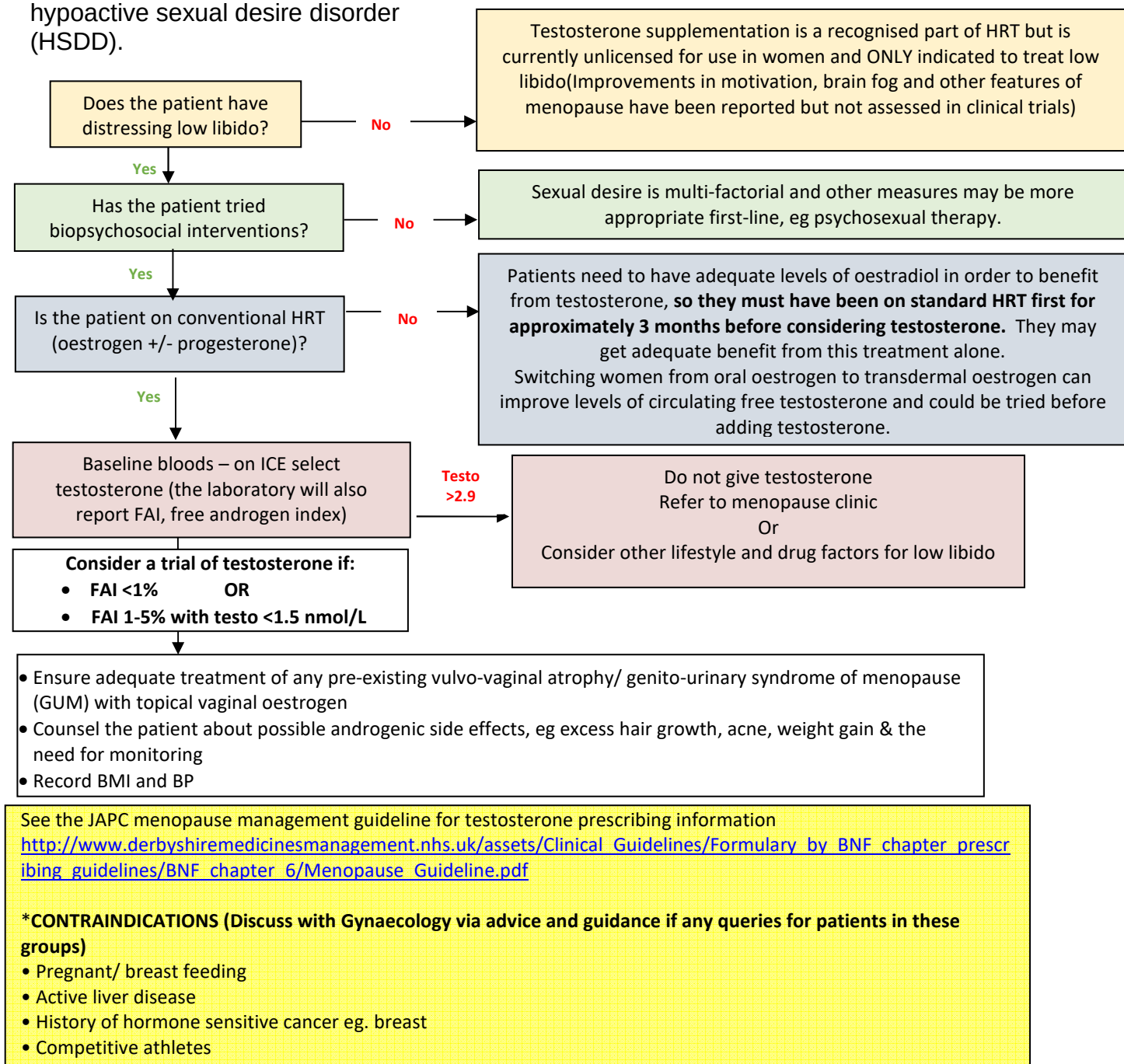
Referral to Gynaecology / Endocrinology would be appropriate for women with POI - for consideration of further genetic tests in non-iatrogenic causes, psychological support, hormone replacement therapy and advice about fertility being the mainstays of specialist involvement.

***FSH**

Note :Lack of rise in FSH would not exclude hypopituitarism. If the clinical context is in line with a potential pituitary problem then a broader set of hormone tests may be warranted- please liaise with Endocrinology (eg via Advice and Guidance section of eRS) if needed'

Testosterone replacement

Testosterone supplementation should only be considered in women who complain of low sexual desire after a biopsychosocial approach has excluded other causes such as relationship, vulvovaginal pain, psychological and medication related (e.g. SSRIs/SNRIs) hypoactive sexual desire disorder (HSDD).



Clinical response to treatment is a better guide than serum levels.

In case of androgenic side effects AT ANY TIME, stop testosterone immediately

Review at 2-3 months

- Systemic changes (acne, hirsutism, male pattern baldness) – **STOP immediately**, check testosterone level and discuss with secondary care via advice and guidance
- **Check testosterone levels** are within the physiologically normal range – **STOP immediately** if testo >2.9 nmol/L OR FAI >5%

Annual monitoring – at least ONCE every 12 months

- **Check testosterone levels** and side-effects as above
- Record BMI, BP

Hormone replacement therapy (HRT)

Hormone tests play no role in deciding the type of HRT for symptomatic menopause. Neither FSH nor oestrogen measurements are recommended in women receiving oral HRT as results are difficult to interpret meaningfully and vary depending on oestrogen type. Most oral oestrogens will not be detected by the laboratory method for measuring serum oestradiol.

The role of serum oestradiol measurement in assessing response to treatment in women using HRT is extremely limited. In almost all cases, decisions about the type and dose of HRT should be based on symptomatic response. Vasomotor symptoms, if present, are a particularly good guide to response - if they persist despite treatment, oestrogen replacement is probably not sufficient; if they have been satisfactorily controlled, further increases in oestrogen dose are less likely to help the patient.

There is no role for oestradiol measurement in perimenopause women. If they are still having menstrual cycles, even if irregular, their native oestrogen production means serum oestradiol levels cannot give any information about the response to HRT.

There is no role for oestradiol measurement in women using HRT with oral oestrogen. Hepatic metabolism means that much of the circulating oestrogen will not be in the form of oestradiol, and will not be detected by the assay.

Even in postmenopausal women using transdermal oestradiol, serum levels may vary from day to day in the same individual. The level needed to adequately control symptoms varies from person to person, and there is no agreed target range.

Where there may be a limited role for serum oestradiol levels is in postmenopausal women who still experience significant symptoms despite being on a higher dose of transdermal oestradiol (75 - 100 mcg patch or equivalent). An oestradiol level lower than 200 pmol/l suggests insufficient absorption, and clinicians may consider changing to a different transdermal product rather than increasing the dose (or considering oral oestradiol, if no contraindication). If serum levels are >400pmol/l consider other causes for symptoms, or other interventions such as commencing testosterone therapy.

Primary care clinicians should consider escalating such complex cases to secondary care via the Advice and Guidance service to gynaecology.

Off-Label use of HRT

In recent years there has been a trend, particularly among private-sector menopause specialists, to prescribe some patients oestrogen doses in excess of those licensed in the UK. The British Menopause Society has stated that "There are no guidelines which recommend that the dose of oestrogen provided should be increased out of license or that different forms of oestrogen should be added together to control symptoms. In the event of a significant event occurring (such as endometrial hyperplasia or cancer) in women using different formulations of oestrogen, outside of license, the prescribing clinicians are responsible for having put patient safety at risk."

If primary care clinicians have concerns about issuing such a prescription, they should discuss with the original prescriber in the first instance, or, if not possible, seek advice from the NHS secondary care provider via the Advice and Guidance service.

Contraceptive Advice

Guidance from the Faculty of Sexual & Reproductive Healthcare for women stopping contraception can be found at <https://www.fsrh.org/standards-and-guidance/documents/fsrh-guidance-contraception-for-women-aged-over-40-years-2017/>

FSH levels offer no secure guide for contraceptive advice.

**In both peri- and post-menopausal patients, future cycles
cannot be excluded even if the FSH is >30 IU/L**

FSH levels are not affected by progesterone-only containing medications (Mirena, POP, Implanon or Depo) and laboratory testing may have some value in determining when contraception can be stopped in these women.

Women over the age of 50 years who are amenorrhoeic and wish to stop progesterone-only contraception can have their FSH levels checked. If the level is ≥ 30 IU/L contraception can be stopped after 1 year.

FSH is not a reliable indicator of ovarian failure in women using combined hormones, even if measured during the hormone-free interval.

In those coming off oral contraception, alternative contraception should be used for one year of amenorrhoea in those aged >50 years and for two years in those aged <50 years.

In women with non-surgical POI there is still a chance of spontaneous ovulation so contraception should be advised if they wish to avoid pregnancy.

Contacts

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Patient Information website

<http://www.patient.co.uk/health/Menopause-and-HRT.htm>

References

The British Menopause Society (BMS). Tools for Clinicians. Testosterone Replacement in Menopause. Dec 2022. <https://thebms.org.uk/publications/tools-for-clinicians/testosterone-replacement-in-menopause/>

NICE Guideline No 23 (NG23). Menopause: Diagnosis and Management. Last updated November 2024. <http://www.nice.org.uk/NG23>

Contraception for women aged over 40 years. Clinical Effectiveness Unit, Faculty of Sexual and Reproductive Healthcare (Sep 2019)Updated July 2023. <https://www.fsrh.org/standards-and-guidance/documents/fsrh-guidance-contraception-for-women-aged-over-40-years-2017/>

Guideline on the management of premature ovarian insufficiency (European Society of Human Reproduction and Embryology, 2024). <https://www.eshre.eu/Guidelines-and-Legal/Guidelines/Management-of-premature-ovarian-insufficiency>

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Secondary Care UHDB	13.06.25	Dr P Hinstridge	Consultant Gynaecologist
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